

Notes de Théorie Algébrique des Nombres

GitHub Repository: [Oxke/appunti/AlgebraSuperiore](https://github.com/Oxke/appunti/AlgebraSuperiore)

1er periode, 2026 - 2027,

Indice

1	Number fields and their rings of integers	3
1.1	Integrality	3
1.2	How to compute rings of integers	5
1.2.1	Theoretical algorithm to compute rings of integer	8
1.3	Algebraic properties	9
1.3.1	Fractional ideals	9
1.3.2	Factorization of ideals	10
1.4	Ramification	10
2	Geometry of numbers	12
2.1	Completions	12
2.1.1	Completion of archimedean valued fields	12
2.1.2	Completion of nonarchimedean valued fields	13
2.1.3	Extensions of complete valued fields	14
2.2	Ramification Theory	16
2.2.1	Complete discrete valued fields	16

Capitolo 1

Number fields and their rings of integers

1.1 Integrality

Definizione 1

Take $A \subseteq B$ an inclusion of domains. An element $x \in B$ is said to be **integral over** A if it is a root of a monic polynomial in $A[X]$.

The ring B is said to be **integral over** A if all elements of B are integral.

Esempio 2. The complex numbers $\overline{\mathbb{Z}}$ that are integral over $\mathbb{Z}$ are called the *algebraic integers*.

Esempio 3. Let $d \neq 0, 1$ squarefree. Then

$$\mathbb{Z}[\sqrt{d}] = \{a + b\sqrt{d} : a, b \in \mathbb{Z}\}$$

is integral over $\mathbb{Z}$. Indeed if $x = a + b\sqrt{d}$, then $(x - a)^2 - b^2d = 0$

Esempio 4. If $\zeta = e^{\frac{2\pi i}{n}}$ then $\zeta^n = 1$, thus $\zeta \in \overline{\mathbb{Z}}$

Proposizione 5. Let A be a UFD. Let $k = \mathcal{Q}(A)$ the fraction field of A . An element $x \in K$ is integral over A if and only if $x \in A$

Dimostrazione. Write $x = \frac{p}{q}$, with $p, q \in A$ and $\text{mcd}(p, q) = 1$. Then in x is integral over A , it satisfies an equation

$$x^d + a_{d-1}x^{d-1} + \cdots + a_0 = 0 \quad a_i \in A \quad \forall i$$

Then

$$\begin{aligned} \left(\frac{p}{q}\right)^d + a_{d-1}\left(\frac{p}{q}\right)^{d-1} + \cdots + a_0 &= 0 \\ \implies p^d + a_{d-1}p^{d-1}q + \cdots + a_0q^d &= 0 \end{aligned}$$

hence $q|p^d$ but they are coprime, hence $q \in A^*$ and thus $x \in A$ □

Conversely, if $x \in A$, it is a root of $X - x$.

Corollario 6. $\mathbb{Q} \cap \overline{\mathbb{Z}} = \mathbb{Z}$

Proposizione 7. Let $A \subseteq B$ inclusion of domains. Fix $b_1, \dots, b_n \in B$ not all zero. Set $P = \sum_{i=1}^n Ab_i$ and take b such that $bP \subseteq P$.

We can write

$$bb_i = \sum_{j=1}^n a_{ij}b_j \quad a_{ij} \in A$$

Then introduce $M = (a_{ij}) \in \mathcal{M}_n(A)$ and let

$$\chi_M(T) = \det(TI_n - M)$$

then $\chi_M(b) = 0$

Osservazione. M^t

Dimostrazione. Set $N = bI_n - M$. We have

$$N \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} = 0$$

So

$$\det N \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} = ({}^t\text{Conj} N) N \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} = 0$$

Hence, since there exists i_0 with $b_{i_0} \neq 0$,

$$\det N = 0 = \chi_M(b)$$

□

Esempio 8. Let $A = \mathbb{Z}$, $B = \mathbb{C}$ If we let $b_1 = 1, b_2 = \sqrt[3]{2}, b_3 = \sqrt[3]{4}$. Then we can take $b = 1 + 3\sqrt[3]{2} - \sqrt[3]{4}$

Corollario 9. Let $A \subseteq B$ inclusion of domains. Take $x_1, \dots, x_r \in B$. Then $x_1, \dots, x_r$ are integral over A if and only if $A[x_1, \dots, x_r]$ is finitely generated as an A -module.

Dimostrazione.

$\Rightarrow$ Assume $x_1, \dots, x_r$ are integral over A . By induction, we may and do assume that $r = 1$. Take $\mu \in A[X]$ monic with $\mu(x_1) = 0$. Take $f(x_1) = x \in A[x_1]$, with $f \in A[X]$. Since μ is monic, we can write

$$f = \mu Q + R$$

with $\deg R < \deg \mu$ and $Q, R \in A[X]$. Finally, $f(x_1) = R(x_1) \in \sum_{i=1}^{\deg \mu - 1} Ax_1^i$ hence $A[x_1]$ is finitely generated.

$\Leftarrow$ Assume $A[x_1, \dots, x_r]$ is finitely generated as an A -module. Let $b_1, \dots, b_n$ be generators and set $b = x_i$. Apply the previous proposition to this setting and get a monic $\chi \in A[X]$ with $\chi(x_i) = 0$. Hence, x_i is integral over A .

□

Corollario 10. Let $A \subseteq B$ inclusion of domains. Let $\overline{A}$ be the set of elements of B that are integral over A then $\overline{A}$ is a subring of B

Dimostrazione. Take $x, y \in \overline{A}$ and consider $C = A[x, y]$. Then C is finitely generated as an A -module. But $C = A[x, y, x+y, xy]$, hence $x+y, xy \in \overline{A}$ □

In particular, $\overline{\mathbb{Z}}$ is a ring.

Corollario 11. Take $A \subseteq B \subseteq C$ a tower of inclusion of domains. Assume C is integral over b and B is integral over A . Then C is integral over A .

Dimostrazione. Take $x \in C$. Let $f \in B[X]$ monic with $f(x) = 0$ let $b_1, \dots, b_n$ be the coefficients of f and set $B' = A[b_1, \dots, b_n] \subseteq B$. Since $b_1, \dots, b_n$ are integral over A , B' is a finitely generated A -module. Since x is integral over B' , $B'[x]$ is a finitely generated B' -module. Hence, $B'[x]$ is finitely generated as A -module, so x is integral over A . $\square$

In particular, if $\overline{A} = \{x \in B : x \text{ integral over } A\}$, then $\{x \in C : x \text{ integral over } A\} = \{x \in C : x \text{ integral over } \overline{A}\}$

Definizione 12

Let $A \subseteq B \subseteq C$ domains.

1. The **integral closure** of A in B is the ring

$$\overline{A} = \{x \in B : x \text{ integral over } A\}$$

2. If $B = \mathcal{Q}(A)$, then $\overline{A}$ is called the **integral closure of A** .
3. If $\overline{A} = A$ we say that A is **integrally closed** or **normal**.
4. If k is a finite extension of $\mathbb{Q}$ (i.e. a number field) then the integral closure of $\mathbb{Z}$ in k is called the **ring of integers** of k and is denoted $\mathcal{O}_k := i(K) \cap \overline{\mathbb{Z}}$ with $i : K \hookrightarrow \mathbb{C}$ injective.

Esempio 13. UFDs are integrally closed. In particular, $\mathcal{O}_{\mathbb{Q}} = \mathbb{Z}$

Esempio 14. If $k = \mathbb{Q}(\sqrt{d})$ with $d \neq 0, 1$ squarefree, then $\mathbb{Z}[\sqrt{d}] \subseteq \mathcal{O}_k$. In general (depending on the value of d), $\mathbb{Z}[\sqrt{d}]$ is not necessarily equal to $\mathcal{O}_k$. If $d \equiv 1 \pmod{4}$ then $\frac{1+\sqrt{d}}{2} \in \mathcal{O}_k \setminus \mathbb{Z}[\sqrt{d}]$

The Reference Framework (RF) is having

- A a domain with $\mathcal{Q}(A) = K$
- L/K a finite separable extension
- $B \subseteq L$ the integral closure of A in L

In particular, we are interested in the cases with $A = \mathbb{Z}$ and $K = \mathbb{Q}$

1.2 How to compute rings of integers

During this section we work with RF.

Proposizione 15. Take $x \in L$. Then $x \in B$ if and only if the minimal polynomial μ of x has coefficient in A .

Dimostrazione.

$\Rightarrow$ Assume $x \in B$. Take $f \in A[X]$ monic with $b(x) = 0$. Then $\mu|f$ in $K[X]$. So the roots $z_1, \dots, z_n$ of μ are roots of f . Hence, $z_1, \dots, z_n$ are integral over A . But the coefficients of μ are in $A[z_1, \dots, z_n]$ hence they are integral over A . But they are in K and A is integrally closed, hence they are in A and $\mu \in A[X]$

$\Leftarrow$ obvious

□

Corollario 16. Take $A = \mathbb{Z}$, $K = \mathbb{Q}$ and $L = \mathbb{Q}(\sqrt{d})$ with $d \neq 0, 1$ squarefree. Then

$$\mathcal{O}_L = \begin{cases} \mathbb{Z}[\sqrt{d}] & d \equiv 2, 3 \pmod{4} \\ \mathbb{Z}\left[\frac{1+\sqrt{d}}{2}\right] & d \equiv 1 \pmod{4} \end{cases}$$

Dimostrazione. Take $a+b\sqrt{d} \in L$, with $a, b \in \mathbb{Q}$ and $b \neq 0$. The minimal polynomial is

$$\mu = (X - a)^2 - b^2 d = X^2 - 2aX + a^2 - db^2$$

Hence $x \in \mathcal{O}_L \iff \mu \in \mathbb{Z}[X] \iff 2a \in \mathbb{Z}$ and $a^2 - db^2 \in \mathbb{Z}$. If $x \in \mathcal{O}_L$, then

$$(2a)^2 - 4(a^2 - db^2) = 4db^2 = d(2b)^2$$

Since d is squarefree, $2b \in \mathbb{Z}$. Write $a = \frac{m}{z}$, $b = \frac{n}{z}$ with $m, n \in \mathbb{Z}$. We conclude in the two cases with the thesis. □

We introduce some algebraic tools. L/K is finite separable. By the primitive element theorem we know $L = K(\alpha)$, with $\alpha \in L$ and $L = K(\alpha) \cong K[X]/(\mu)$ where μ is the minimal polynomial of α .

Take L^s a separable closure of L . $\text{Hom}_K(L, L^s) = \{\sigma : L \hookrightarrow L^s : \sigma|_K = \text{id}_K\} \cong \{\text{roots of } \mu \text{ in } L^s\}$. In particular, $|\text{Hom}_K(L, L^s)| = [L : K]$. Coming back to the question of finding out if some elements of L are integral or not over A , we know how to compute polynomials that vanish at some $x \in L$:

$$\begin{aligned} m_x : L &\longrightarrow L \\ y &\longmapsto m_x(y) = xy \end{aligned}$$

and $\chi_{m_x}(x) = 0$

Definizione 17

$\chi_{m_x} =: \chi_x$ is called the characteristic polynomial of x .

$$\chi_x = X^n - a_{n-1}X^{n-1} + \cdots + (-1)^n a_0$$

and $a_{n-1} = \text{Tr}_{L/K}(x)$ the trace of x , $a_0 = N_{L/K}(x)$ the norm of x .

Proposizione 18. We have: μ_x minimal polynomial of x . Then

$$\chi_x = (\mu_x)^{[LK(x)]} = \prod_{\sigma \in \text{Hom}_K(L, L^s)} (X - \sigma(x))$$

and

$$\text{Tr}_{L/K}(x) = \sum_{\sigma \in \text{Hom}_K(L, L^s)} \sigma(x) \quad ; \quad N_{L/K}(x) = \prod_{\sigma \in \text{Hom}_K(L, L^s)} \sigma(x)$$

Dimostrazione.

$$\begin{pmatrix} 0 & 0 & \cdots & 0 & -a_{n-1} \\ 1 & 0 & \cdots & 0 & -a_{n-2} \\ 0 & 1 & \cdots & \vdots & \vdots \\ \vdots & \vdots & \ddots & 0 & \vdots \\ 0 & 0 & \cdots & 1 & -a_0 \end{pmatrix}$$

where $\mu = X^n + a_{n-1}X^{n-1} + \dots + a_0$. Since the characteristic poly of tthis matrix is μ , we get

$$\chi_x = \mu_x^v = \mu_x^{[L:K(x)]}$$

We know:

$$\begin{aligned}\mu_x &= \prod_{\tau \in \text{Hom}_K(K(x), L^s)} (X - \tau(x)) \\ \chi_x &= \prod_{\tau \in \text{Hom}_K(K(x), L^s)} (X - \tau(x))^{[L:K(x)]}\end{aligned}$$

We have to check that, for each $\tau \in \text{Hom}_K(K(x), L^s)$, there exists exactly $[L : K(x)]$ elements $\sigma \in \text{Hom}_K(L, L^s)$ with $\sigma|_{K(x)} = \tau$.

Then, by PET, $L = K(x, y)$ with some $y \in L$. Let μ_y be the minimal polynomial of y over $K(x)$. Fixing $\tau \in \text{Hom}_K(K(x), L^s)$,

$$\text{Hom}_{K(x)}(L, L^s) = \{\sigma : L \hookrightarrow L^s : \sigma|_{K(x)} = \tau\} \text{ has } [L : K(x)] = \deg \mu_y \text{ elements}$$

□

Corollario 19. Take $x \in L$. Then $x \in B$ if and only if $\chi_x \in A[X]$. In particular, if $x \in B$, then $\text{Tr}_{L/K}(x) \in A$ and $N_{L/K}(x) \in A$

Dimostrazione.

□

Fix $\mathbf{x} = (x_1, \dots, x_n)$ a K -basis of L . Given $x = \sum_{i=1}^n d_i x_i$, $d_i \in K$, I want to express in terms of the d_i 's whether $x \in B$ or $x \notin B$.

Lemma 20. $\mathcal{Q}(B) = L$. Better, for each $x \in L$, there exists $y \in A \setminus \{0\}$ such that $yx \in B$.

Dimostrazione. Take $\mu_x \in K[X]$ the minimal polynomial of x . Write $a_i = b_i/c_i$, with $b_i, c_i \in A$. Set $c := (c_0, \dots, c_{m-1})$ and $z := cx$. Then

$$\left(\frac{z}{c}\right)^m + a_{m-1}\left(\frac{z}{c}\right)^{m-1} + \dots + a_0 = 0$$

by multiplying everything times c^m we find $z \in B$, hence $x = \frac{z}{c}$ with $z \in B$ and $c \in A$. □

Up to multiplying $\mathbf{x}$ by some well-chosen $a \in A$, we may assume $x_1, \dots, x_n \in B$. We come back to

$$x = \sum_{i=1}^n \lambda_i x_i, \quad \lambda_i \in K$$

If $x \in B$, then $\forall i, x x_i \in B_n$ and $\text{Tr}_{L/K}(x x_i) = \sum_{j=1}^n \lambda_j \text{Tr}_{L/K}(x_i x_j) \in A$.

If we set $M_{\mathbf{x}} = (\text{Tr}_{L/K}(x_i x_j)) \in \mathcal{M}_n(A)$, then $M_{\mathbf{x}} \begin{pmatrix} \lambda_1 \\ \lambda_2 \\ \vdots \\ \lambda_n \end{pmatrix} \in A^n$ and hence its

determinant is in A^n .

Proposizione 21. If $d(\mathbf{x}) = \det(M_{\mathbf{x}}) = \det(\text{Tr}_{L/K}(x_i x_j))$ and $d(\mathbf{x})B \subseteq \sum_{i=1}^n A x_i$

Definizione 22: discriminant

$d\mathbf{x}$ is called the **discriminant** of $\mathbf{x}$

Computation of the discriminant

$$\begin{aligned} d(\mathbf{x}) &= \det(\mathrm{Tr}_{L/K}(x_i x_j)) = \\ &= \det\left(\sum_{\sigma \in \mathrm{Hom}_K(L, L^s)} \sigma(x_i) \sigma(x_j)\right) = \\ &= \left(\det(\sigma(x_j))_{\sigma, j}\right)^2 \end{aligned}$$

1.2.1 Theoretical algorithm to compute rings of integer

Take $A = \mathbb{Z}$, $K = \mathbb{Q}$, L a number field and $B = \mathcal{O}_L$, then

- 1) Find a $\mathbb{Q}$ -basis $\mathbf{x} = (x_1, \dots, x_n)$ of L contained in B e.g. $\mathbf{x} = (1, x, x^2, \dots, x^{n-1})$ with $x \in B$ a primitive element for $L/\mathbb{Q}$
- 2) Compute $d(\mathbf{x})$
- 3) Say that

$$\bigoplus_i \mathbb{Z}x_i \subseteq B \subseteq \frac{1}{d(\mathbf{x})} \bigoplus_i \mathbb{Z}x_i$$

- 4) Compute representation $y_1, \dots, y_n$ of the quotient

$$\left(\frac{1}{d(\mathbf{x})} \bigoplus_i \mathbb{Z}x_i\right) / \left(\bigoplus_i \mathbb{Z}x_i\right)$$

Proposizione 23. *In the RF. Assume A is a PID. Then*

1. *There exists a K -basis $\mathbf{x} = (x_1, \dots, x_n)$ of L contained in B such that*

$$B = \bigoplus_i Ax_i$$

*such a basis $\mathbf{x}$ is called an **integral basis** of B*

2. *Assume further that $A = \mathbb{Z}$. Then $d(\mathbf{x})$ is independant of the choice of an integral basis $\mathbf{x}$ and if $\mathbf{y}$ is any K -basis of L contained in B , then*

$$d(\mathbf{y}) = [B : \Gamma_{\mathbf{y}}]^2 \cdot d(\mathbf{x})$$

Osservazione. If $A = \mathbb{Z}$ and $K = \mathbb{Q}$. Say you have a $\mathbb{Q}$ -basis $\mathbf{y}$ of L contained in $\mathcal{O}_L$ such that $d(\mathbf{y})$ is squarefree. Then $\mathcal{O}_L = \bigoplus \mathbb{Z}y_i$

Dimostrazione. 1) Take $\mathbf{y}$ a K -basis of L contained in B . We know

$$\bigoplus_i Ay_i \subseteq B \subseteq \frac{1}{d(\mathbf{y})} \bigoplus_i Ay_i$$

By classication of finitely generated modules over PIDs, B is a free A -module of rank $[L : K]$. Hence, there exists $\mathbf{x} = (x_1, \dots, x_n)$ such that $B = \bigoplus_i Ax_i$. Necessarily, $\mathbf{x}$ is also a K -basis of L

- 2) Let $\mathbf{x}$ be an integral basis of $B = \mathcal{O}_L$. Let $\mathbf{y}$ be any $\mathbb{Q}$ -basis of L contained in $\mathcal{O}_L$. We know that $d(\mathbf{x})$ (resp. $d(\mathbf{y})$) is the determinant of the trace form $\mathrm{Tr}_{L/K}$ on the basis $\mathbf{x}$ (resp. $\mathbf{y}$). If P is the transition matrix from $\mathbf{x}$ to $\mathbf{y}$, then $d(\mathbf{y}) = d(\mathbf{x})(\det P)^2$. By Smith's nor form theorem (théorème de la base adaptée), $|\det P| = [\bigoplus_i \mathbb{Z}x_i : \bigoplus_i \mathbb{Z}y_i] = [\mathcal{O}_L : \Gamma_{\mathbf{y}}]$. We conclude with the equivalence in the thesis. Moreover, if $\mathbf{y}$ was an integral basis, then $\Gamma_{\mathbf{y}} = \mathcal{O}_L$ and thus $d(\mathbf{y}) = d(\mathbf{x})$

□

Definizione 24

Let L be a number field. The discriminant of an integral basis of L is called **discriminant** of L and it's noted Δ_L

1.3 Algebraic properties

Rings of integers do not satisfy many properties. They are not PID generally for instance.

Definizione 25: Dedekind domains

A domain A is said to be a **Dedekind domain** if it is Noetherian, integrally closed and every non-zero prime ideal in A is maximal. Equivalently, it's a Noetherian domain with Krull dimension at most 1.

Esempio 26. PIDs are Dedekind domains. UFDs are not all Dedekind domains. For instance, in $A = \mathbb{C}[x, y]$ (x) is not maximal.

Teorema 27

Working in the Framework RF (even though separability of L/K is not needed). Assume A is a Dedekind domain. Then B is also a Dedekind domain.

Osservazione. In particular, rings of integers in number fields are Dedekind domains.

Dimostrazione. (We use the assumption of separability of L/K). We already know that B is integrally closed. We know that, if $\mathbf{x} = (x_1, \dots, x_n)$ is a K -basis of L contained in B and $B \subseteq \frac{1}{d(\mathbf{x})} \bigoplus A x_i$ which is a finitely generated A -module, hence noetherian. It follows that B is a noetherian ring.

Take a non-zero prime ideal $\mathfrak{p}$ in B . Set $\mathfrak{q} = \mathfrak{p}^c = \mathfrak{p} \cap A$. Then $\mathfrak{q}$ is a prime ideal in A . Moreover, if one takes $x \in \mathfrak{p} \setminus \{0\}$, then x satisfies some equation

$$x^n + a_{n-1}x^{n-1} + \dots + a_0 = 0 \quad a_i \in A, a_0 \neq 0$$

Hence

$$a_0 = -x^n - a_{n-1}x^{n-1} - \dots - a_1x \in \mathfrak{p} \cap A = \mathfrak{q}$$

thus $\mathfrak{q}$ is non-zero, hence maximal. $B/\mathfrak{p}$ is a domain which is also a finite-dimensional $(A/\mathfrak{q})$ -vector space (B is finitely generated A -module). Hence, $B/\mathfrak{p}$ is a field and $\mathfrak{p}$ is maximal. $\square$

1.3.1 Fractional ideals

Definizione 28

Let A be a domain with $\mathcal{Q}(A) = K$. A fractional ideal I in K is an A -submodule of K such that there exists $a \in A \setminus \{0\}$ satisfying $aI \subseteq A$

Esempio 29. If A is a PID, fractional ideals are the xA with $x \in K \setminus \{0\}$. In particular if A is not a field, K is not a fractional ideal.

Teorema 30

Let A be a Dedekind domain. All non-zero prime ideals in A are invertible.

Take $\mathfrak{p}$ a non-zero prime ideal in A . If J is a fractional ideal such that $\mathfrak{p}J \subseteq A$, then we define

$$J \subseteq \{x \in K : x\mathfrak{p} \subseteq A\} =: \mathfrak{p}^{-1}$$

Note that if $y \in \mathfrak{p} \setminus \{0\}$, then $y\mathfrak{p}^{-1} \subseteq A$, hence $\mathfrak{p}^{-1}$ is a fractional ideal.

The theorem will then follow from the following

Proposizione 31. *Let A be a Dedekind domain, $\mathfrak{p}$ a non-zero prime ideal in A and I any non-zero ideal in A and I any non-zero ideal in A . Then $I \not\subseteq \mathfrak{p}^{-1}I$. In particular, $\mathfrak{p}^{-1}\mathfrak{p} = A$*

Osservazione. In particular, $\mathfrak{p} \subsetneq \mathfrak{p}^{-1}\mathfrak{p} \subseteq A$ since $\mathfrak{p}$ is maximal, $\mathfrak{p}^{-1}\mathfrak{p} = A$

Dimostrazione. Step 1: We prove that every nonzero ideal $I \subseteq A$ contains a product of nonzero prime ideals. Assume by contradiction this is wrong. Consider $\mathcal{E}$ a set of nonzero ideals $I \subseteq A$ that do not contain a product of nonzero prime ideals. By assumption $\mathcal{E} \neq \emptyset$ and using Notherianity, it has a maximal element I_0 . I_0 is not prime, so we can find $a, b \in A \setminus I_0$ such that $ab \in I_0$. Then $I_0 + (a)$ and $I_0 + (b)$ both contain product of nonzero prime ideals. Hence, $(I_0 + (a))(I_0 + (b)) \subseteq I_0$ contains a product of nonzero prime ideals, which leads to a contradiction.

Step 2: We prove that $A \subsetneq \mathfrak{p}^{-1}$. $\supseteq$ is obvious. Take now $a \in \mathfrak{p} \setminus \{0\}$; by *Step 1* we can write $\mathfrak{p}_1 \dots \mathfrak{p}_r \subseteq (a) \subseteq \mathfrak{p}$ and we suppose r as small as possible, with all $\mathfrak{p}_i$ prime ideals. There exists i_0 such that $\mathfrak{p}_{i_0} \subseteq \mathfrak{p}$ and we may assume $i_0 = r$. Since $\mathfrak{p}_r$ is maximal, $\mathfrak{p}_r = \mathfrak{p}$. Moreover, since r is as small as possible, we can find $b \in \mathfrak{p}_1 \dots \mathfrak{p}_{r-1} \setminus (a)$. Then $a^{-1}b \notin A$ and $\forall x \in \mathfrak{p}$, $a^{-1}bx \in a^{-1}\mathfrak{p}_1 \dots \mathfrak{p}_{r-1}\mathfrak{p} = a^{-1}\mathfrak{p}_1 \dots \mathfrak{p}_r \subseteq a^{-1}(a) = A$. Hence, $a^{-1}b \in \mathfrak{p}^{-1}$.

Step 3: We prove the proposition $\mathfrak{p}^{-1}I \not\supseteq I$ by contradiction. Assume $\mathfrak{p}^{-1}I = I$ and take $x \in \mathfrak{p}^{-1}$. Then $x \in \mathfrak{p}^{-1}$. Then $xI \subseteq I$ and by Notherianity of A , I is finitely generated. By a previous proposition, this allows to find a monic $\chi \in A[X]$ such that $\chi(x) = 0$. But A is integrally closed, hence $x \in A$. Hence $\mathfrak{p}^{-1} = A$ which contradicts *Step 2*. $\square$

1.3.2 Factorization of ideals

Teorema 32

Let A be a Dedekind domain. Then every nonzero ideal of A can be written as a product of nonzero prime ideals, in a unique way up to the order.

Dimostrazione. Existence: Let $\mathcal{E}$ be the family of ideals in A other than (0) and (1) that are not product of prime ideals. By contradiction, assume $\mathcal{E} \neq \emptyset$. By Notherianity, we can find a maximal element $I_0 \in \mathcal{E}$. Since $I_0 \neq A$ we can find a maximal ideal $\mathfrak{p}$ of A containing I_0 . Set $J = \mathfrak{p}^{-1}I_0$.

Then

$$i_0 \subsetneq \mathfrak{p}^{-1}I_0 \subseteq \mathfrak{p}^{-1}\mathfrak{p} = A$$

So $\mathfrak{p}^{-1}I_0$ is an ideal of A not in $\mathcal{E}$. So $\mathfrak{p}^{-1}I_0 = \mathfrak{p}_1 \dots \mathfrak{p}_r$ for some prime ideals $\mathfrak{p}_1, \dots, \mathfrak{p}_r$. We conclude $I_0 = \mathfrak{p}\mathfrak{p}^{-1}I_0 = \mathfrak{p}\mathfrak{p}_1 \dots \mathfrak{p}_r$. $\square$

1.4 Ramification

Definizione 33

In the RF with A Dedekind, take $\mathfrak{p}$ a prime ideal in A . Write the factorization of $\mathfrak{p}B = p_1^{e_1} \dots p_g^{e_g}$.

1.

Proposizione 34. *Almost all prime ideals of A are unramified in L .*

Dimostrazione. Let $\alpha \in B$ a primitive element for L/K and take μ its minimal polynomial. Set $\square$

Teorema 35

Consider the *discriminant ideal* $\mathfrak{d}_{B/A}$ of A generated by the discriminants of all K -bases of L contained in B .

A prime ideal $\mathfrak{p}$ of A is ramified in $L \iff \mathfrak{p} | \mathfrak{d}_{B/A}$

Capitolo 2

Geometry of numbers

2.1 Completions

Let $(K, |\cdot|)$ a valued field and let R be the ring of all Cauchy sequences in K . let $\mathfrak{m} \subseteq R$ be the ideal of sequences converging to 0.

Lemma 36. $\mathfrak{m}$ is a maximal ideal in R

Dimostrazione. Take $(a_n)_n \in R \setminus \mathfrak{m}$. Since $(a_n)_n \notin \mathfrak{m}$, there exists $\varepsilon > 0$ and $N \geq 0$ such that $\forall n \geq N, |a_n| \geq \varepsilon$. Now take $(b_n)_n$ a sequence such that $\forall n \geq N, a_n b_n = 1$. Then $(b_n)_n \in R$ and hence $(b_n)_n$ is an inverse for $(a_n)_n$ in $R/\mathfrak{m}$. So $(R/\mathfrak{m})$ is a field and $\mathfrak{m}$ is maximal. $\square$

Definizione 37

The **completion** of $(K, |\cdot|)$ is the field $\widehat{K} = R/\mathfrak{m}$. K is embedded in $\widehat{K}$ by sending $a \in K$ to the sequence that is constant equal to a .

Take $(a_n)_n \in R$. We know $||a_m| - |a_n|| \leq |a_m - a_n|$ so $(|a_n|)_n \subseteq \mathbb{R}$ is Cauchy. Let $|(a_n)_n|$ be its limit. If we take $(b_n)_n \in R$ with $(a_n)_n - (b_n)_n \in \mathfrak{m}$, then $||a_n| - |b_n|| \leq |a_n - b_n|$, so $(|a_n|)_n$ and $(|b_n|)_n$ have the same limit. Hence $(a_m)_n \rightarrow |(a_n)_n|$ is an absolute value of $\widehat{K}$ that extends the absolute value of K . If $(a_n)_n \in R$, then $a_n \rightarrow (a_n)_n$ in $\widehat{K}$.

Proposizione 38. $(\widehat{K}, |\cdot|)$ is complete.

Osservazione. Let $(\widehat{K}', |\cdot|')$ be another complete value field such that $|\cdot|'$ extends the absolute value of K and K is dense in $\widehat{K}'$ and $\widehat{K}'$ is complete. Then there exists a unique K -isomorphism $\sigma : \widehat{K} \rightarrow \widehat{K}'$ such that $|\cdot|' \circ \sigma = |\cdot|$.

2.1.1 Completion of archimedean valued fields

Teorema 39: Ostrowski 2

Let $(K, |\cdot|)$ be an archimedean complete valued field. There is an isomorphism $\sigma : K \rightarrow \mathbb{R}$ or $K \rightarrow \mathbb{C}$ such that

$$|\cdot| = (|\cdot|_{\circ} \circ \sigma)^s$$

for some $\mathbb{R} \ni s \geq 1$

2.1.2 Completion of nonarchimedean valued fields

Proposizione 40. *Let $(K, |\cdot|)$ be a valued field with discrete normalized absolute field. Then*

1. *The valuation on $\widehat{K}$ is still discrete normalized.*
2. *Let $\mathcal{O}$ and $\widehat{\mathcal{O}}$ valuation rings of K and $\widehat{K}$ and $\mathfrak{p}$ and $\widehat{\mathfrak{p}}$ the maximal ideals. Then*

$$\forall n \geq 1, \quad \mathcal{O}/\mathfrak{p}^n \cong \widehat{\mathcal{O}}/\widehat{\mathfrak{p}}^n$$

Moreover the diagonal map

$$\begin{aligned} \partial : \widehat{\mathcal{O}} &\longrightarrow \varprojlim_{n \geq 1} \mathcal{O}/\mathfrak{p}^n \\ a &\longmapsto (a \bmod \mathfrak{p}, a \bmod \mathfrak{p}^2, \dots) \end{aligned}$$

is an isomorphism and a homeomorphism (with $\mathcal{O}/\mathfrak{p}^n$ endowed with discrete topology).

3. *Let $\mathcal{R} \subseteq \mathcal{O}$ be a system of representatives of the residue field $k := \mathcal{O}/\mathfrak{p}$ containing 0. Let π be a uniformizer of K . Then every element $x \in \widehat{\mathcal{O}}$ can be uniquely written as*

$$x = \sum_{n=0}^{\infty} a_n \pi^n$$

with $a_0, a_1, \dots \in \mathcal{R}$

Dimostrazione. 1. Take $x \in \widehat{K} \setminus \{0\}$. We can find $y \in K$ with $|x - y| < |x|$. Then $|y| = |x|$. Hence $v(\widehat{K}^\times) = v(K^\times)$

2. Consider the inclusion $i : \mathcal{O} \rightarrow \widehat{\mathcal{O}}$. We have

$$\mathfrak{p}^n = \{x \in \mathcal{O} : v(x) \geq n\} = \{x \in \widehat{\mathcal{O}} : v(x) \geq n\} \cap \mathcal{O} = \widehat{\mathfrak{p}}^n \cap \mathcal{O}$$

So i induces an injective morphism $\widehat{i} : \mathcal{O}/\mathfrak{p}^n \rightarrow \widehat{\mathcal{O}}/\widehat{\mathfrak{p}}^n$. For surjectivity, take $y \in \widehat{\mathcal{O}}$. Since K is dense in $\widehat{K}$, we can find $(y_m)_m$ with values in K such that $y_m \rightarrow y$. $y + \widehat{\mathfrak{p}}^n$ is open. So $y_m \in y + \widehat{\mathfrak{p}}^n$ for $m \gg 0$. So $\widehat{i}(y_m) = \widehat{i}(y)$ for $m \gg 0$ and $\widehat{i}$ is an isomorphism.

Consider the map of the statement ∂ . Then $\ker \partial = \bigcap_{n \geq 0} \widehat{\mathfrak{p}}^n = v^{-1}(\infty) = 0$. For surjectivity, take $(\bar{a}_n)_n$ in the limit. Let a_n be a lift of $\bar{a}_n$ to $\mathcal{O}$ for each n . We have $v(a_m - a_n) \geq \min\{m, n\}$, hence $(a_n)_n$ is Cauchy. Let a be its limit in $\widehat{K}$. Since $a_n \in \mathcal{O}$ for all n , $a \in \widehat{\mathcal{O}}$. The inequality $v(a_m - a_n) \geq \min\{m, n\}$ gives $v(a - a_n) \geq n$ for each n , hence $\partial(a) = (\bar{a}_n)_n$, so ∂ is an isomorphism.

For the *homeo*, $\mathfrak{p}^n$ form a basis of open neighborhoods of 0 in $\widehat{K}$. Then

$$\partial(\mathfrak{p}^n) = \ker \left(\varprojlim_n \mathcal{O}/\mathfrak{p}^n \xrightarrow{\text{proj}} \mathcal{O}/\mathfrak{p}^n \right) = \{(\bar{0}, \dots, \bar{0}, \bar{a}_{n+1}, \dots) \in \varprojlim_n \mathcal{O}/\mathfrak{p}^n\}$$

which form a basis of open neighborhoods of 0 in the inverse limit, hence ∂ is a homeomorphism.

3. We inductively construct a sequence $(a_n)_n$ with values in $\mathcal{R}$ such that $\forall m$, $v(x - \sum_{i=0}^m a_i \pi^i) \geq m + 1$. For $m = 0$, we take $a_0 \in \mathcal{R}$ such that $a_0 \equiv x \bmod \widehat{\mathfrak{p}}$. Now assume a_m is already constructed. Then

$$v \left(x - \sum_{i=0}^m a_i \pi^i \right) \geq m + 1 \implies x - \sum_{i=0}^m a_i \pi^i = b \pi^{m+1} \quad \text{with } b \in \widehat{\mathcal{O}}$$

Choose $a_{m+1} \in \mathcal{R}$ such that $a_{m+1} \equiv b \pmod{\mathfrak{p}}$. Then

$$v\left(x - \sum_{i=0}^{m+1} a_i \pi^i\right) = v(\pi^{m+1}(b - a_{m+1})) \geq m+2$$

which finishes the induction.

Then $x = \sum_{i=0}^{\infty} a_i \pi^i$. Take $b_i \in \mathcal{R}$ such that $x = \sum_{i=0}^{\infty} b_i \pi^i$ and assume $(b_i)_i \neq (a_i)_i$. Take i_0 minimal index with $b_{i_0} \neq a_{i_0}$. Then

$$0 = (a_{i_0} - b_{i_0})\pi^{i_0} + \sum_{i \geq i_0+1} (a_i - b_i)\pi^i$$

where the first addend has valuation i_0 and the second has valuation $\geq i_0 + 1$. This is a contradiction and thus $(a_n)_n$ is unique. $\square$

Definizione 41

Given p prime, the completion of $(\mathbb{Q}, |\cdot|_{v_p})$ is denoted $\mathbb{Q}_p$ and is called the **field of p -adic numbers**. Its valuation ring is denoted $\mathbb{Z}_p$ and is called the **ring of p -adic integers**. $p\mathbb{Z}_p$ is the unique maximal ideal of $\mathbb{Z}_p$ and for all n ,

$$\mathbb{Z}_p \cong \varprojlim_n \mathbb{Z}/p^n \mathbb{Z} \quad \frac{\mathbb{Z}_p}{p^n \mathbb{Z}_p} \cong \frac{\mathbb{Z}}{p^n \mathbb{Z}}$$

and an element $x \in \mathbb{Z}_p$ can be uniquely written as $x = \sum_{i=0}^{\infty} a_i p^i$, with $a_i \in \{0, \dots, p-1\}$

Lemma 42: Hensel

Let $(K, |\cdot|)$ be a complete nonarchimedean valued field with valuation ring $\mathcal{O}$. Take $f \in \mathcal{O}[X]$ a polynomial and $a \in \mathcal{O}$ such that $f'(a) \neq 0$ and $c = \frac{|f(a)|}{|f'(a)|^2} < 1$. Consider the sequence $(a_n)_n$ defined as

$$\begin{cases} a_0 = a \\ a_{n+1} = a_n - \frac{f(a_n)}{f'(a_n)} \end{cases}$$

This sequence is well-defined and converges to a root b of f in $\mathcal{O}$ such that $|b - a| \leq c < 1$

Esercizio 43 2.2.10

2.1.3 Extensions of complete valued fields

Definizione 44

Let $(K, |\cdot|)$ be a valued field. Let V be a finite dimensional K -vector space. A norm $\|\cdot\|$ on V is a map $\|\cdot\| : V \rightarrow \mathbb{R}_{\geq 0}$ such that $\forall x \in V$, $\|x\| = 0 \iff x = 0$, it is homogeneous and sublinear.

Another norm $\|\cdot\|'$ is equivalent to $\|\cdot\|$ if there exist constants $C_1, C_2 \in \mathbb{R}$ such that $C_1 \|\cdot\|' \leq \|\cdot\| \leq C_2 \|\cdot\|'$

Proposizione 45. *If $\dim V < \infty$ and K is complete, then all norms on V are equivalent. In particular, whatever the norm on V , V is complete.*

Dimostrazione. Seen a lot of times, just use $\|\cdot\|_1$. □

Teorema 46

Let $(K, |\cdot|)$ be a complete valued field. Let L be a finite extension. There exists a unique extension $|\cdot|_L$ of $|\cdot|$. For all $x \in L$,

$$|x|_L = |N_{L/K}(x)|^{1/[L:K]}$$

for all $x \in L$ and $(L, |\cdot|_L)$ is complete.

Osservazione. The archimedean case immediately follows from Ostrowski's theorem.

Proof (assuming $(K, |\cdot|)$ discrete valued field and L/K separate). First let's prove **existence**. Let v be a valuation on K . The valuation ring $\mathcal{O}$ of K is a PID. So its integral closure B in L is a Dedekind domain. We moreover know that B contains a prime ideal $\mathfrak{q}$ lying above the maximal ideal $\mathfrak{p}$ of $\mathcal{O}$. We have the $\mathfrak{q}$ -adic valuation $v_{\mathfrak{q}}$ on L , which satisfies $v_{\mathfrak{q}}|_K = e(\mathfrak{q}|\mathfrak{p})v$ so $\frac{v_{\mathfrak{q}}}{e(\mathfrak{q}|\mathfrak{p})}$ is a discrete complete valuation on L that extends v .

For the **uniqueness**, let $|\cdot|_1$ and $|\cdot|_2$ be two absolute values on L extending $|\cdot|$. Then $(L, |\cdot|_1)$ and $(L, |\cdot|_2)$ are normed spaces over $(K, |\cdot|)$. Since $(K, |\cdot|)$ is complete and L/K are finite, $|\cdot|_1$ and $|\cdot|_2$ are equivalent *norms*, hence equivalent absolute values. So $|\cdot|_1 = |\cdot|_2^s$ for some $s \geq 0$. Since $|\cdot|_1|_K = |\cdot|_K$, then $s = 1$, hence $|\cdot|_1 = |\cdot|_2$.

Let's call $|\cdot|_L$ the unique extension of $|\cdot|$ to L . We want to prove the equality in the thesis. Let M/K be the Galois closure of L/K , let $|\cdot|_M$ be the unique extension of $|\cdot|$ to M . For $\sigma \in \text{Gal}(M/K)$, since $|\cdot|_M$ is the *only* extension of $|\cdot|$ to M , then $|\cdot|_M \circ \sigma = |\cdot|_M$. So, for $x \in L$,

$$|x|_L = \left(|x|_L^{[L:K]} \right)^{1/[L:K]} = \left(\prod_{\sigma \in \text{Hom}_K(L, M)} |\sigma(x)|_L \right)^{1/[L:K]} = |N_{L/K}(x)|^{1/[L:K]}$$

□

Corollario 47. Let $(K, |\cdot|)$ be a valued field with completion $(\widehat{K}, |\cdot|_{\widehat{K}})$. Let Ω be an algebraic closure of $\widehat{K}$. Let L/K be a finite extension. Then

1. The absolute value $|\cdot|_{\widehat{K}}$ has a unique extension to Ω , $|\cdot|_{\Omega}$
2. The absolute values on L extending $|\cdot|$ are the $|\cdot|_{\Omega} \circ \sigma$, with $\sigma \in \text{Hom}_K(L, \Omega)$
3. For $\sigma, \tau \in \text{Hom}_K(L, \Omega)$, we have $|\cdot|_{\Omega} \circ \sigma = |\cdot|_{\Omega} \circ \tau$ if and only if there exists $\rho \in \text{Gal}(\Omega/\widehat{K})$ such that $\tau = \rho \circ \sigma$

Dimostrazione.

1. For each finite subextension $M/\widehat{K}$ of $\Omega/\widehat{K}$, the absolute value $|\cdot|_{\widehat{K}}$ has a unique extension $|\cdot|_M$ to M . If $M \subseteq N$ are two finite extensions of $\widehat{K}$ contained in Ω , then

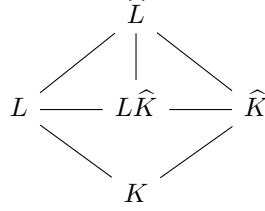
$$|\cdot|_N|_M = |\cdot|_M$$

This allows to define an absolute value $|\cdot|_{\Omega}$ on Ω extending $|\cdot|_{\widehat{K}}$. If $|\cdot|'_{\Omega}$ is another one, then for each $\Omega/M/\widehat{K}$ finite,

$$|\cdot|'_{\Omega}|_M = |\cdot|_M = |\cdot|_{\Omega}|_M$$

So $|\cdot|'_{\Omega} = |\cdot|_{\Omega}$

2. Let $|\cdot|_L$ be some extension of $|\cdot|$ on L . Let $(\widehat{L}, |\cdot|_{\widehat{L}})$ be the completion of $(L, |\cdot|_L)$. The completion $\widehat{K}$ of K identifies with the closure of K in $\widehat{L}$.



We know that $L\widehat{K}$ is complete (finite dimensional over $\widehat{K}$) and contains L as a dense subfield, hence $L\widehat{K} = \widehat{L}$.

So $\widehat{L}/\widehat{K}$ is finite and we can find $\sigma \in \text{Hom}_{\widehat{K}(\widehat{L}, \Omega)}$ such that $|\cdot|_{\widehat{L}}$ and $|\cdot|_{\Omega} \circ \sigma$ are absolute values on $\widehat{L}$ extending $|\cdot|_{\widehat{K}}$. So

$$|\cdot|_{\widehat{L}} = |\cdot|_{\Omega} \circ \sigma \quad \text{and} \quad |\cdot|_L = |\cdot|_{\Omega} \circ \sigma|_L$$

3. Assume $\sigma = \rho \circ \tau$, with $\rho \in \text{Aut}(\Omega/\widehat{K})$. Then $|\cdot|_{\Omega} \circ \sigma = |\cdot|_{\Omega} \circ \rho \circ \tau$. But $|\cdot|_{\Omega} \circ \rho$ is an absolute value on Ω . So, by 1), $|\cdot|_{\Omega} \circ \rho = |\cdot|_{\Omega}$. So $|\cdot|_{\Omega} \circ \sigma = |\cdot|_{\Omega} \circ \tau$. Conversely, assume $|\cdot|_{\Omega} \circ \sigma = |\cdot|_{\Omega} \circ \tau$. Then set $\rho_0 := \tau \circ \sigma^{-1} : \sigma(L) \rightarrow \tau(L)$. We claim that ρ_0 can be extended to some $\widehat{\rho}_0 : \sigma(L)\widehat{K} \rightarrow \tau(L)\widehat{K}$. To do so, take $x \in \sigma(L)\widehat{K} \rightarrow \tau(L)\widehat{K}$ continuously. To do so, take $x \in \sigma(L)\widehat{K}$ and choose $(x_n)_n$ a sequence in $\sigma(L)$ converging to x . For $m, n \geq 1$,

$$\begin{aligned}
 |\rho_0(x_m) - \rho_0(x_n)|_{\Omega} &= |\tau \circ \sigma^{-1}(x_m) - \tau \circ \sigma^{-1}(x_n)|_{\Omega} = |\tau \circ \sigma^{-1}(x_m - x_n)|_W = \\
 &= |\sigma \circ \sigma^{-1}(x_m - x_n)|_{\Omega} = |x_m - x_n|_{\Omega}
 \end{aligned}$$

Hence $(\rho_0(x_n))_n$ is Cauchy, hence it has a limit $\rho_0(x)$ in $\sigma(L)(\widehat{K})$ (which is complete).

One then checks that $\widehat{\rho}_0(x)$ only depends on x and that $\widehat{\rho}_0$ is a $\widehat{K}$ -isomorphism (exercise). We end by extending $\widehat{\rho}_0$ to some $\rho \in \text{Aut}(\Omega/\widehat{K})$. Then we can conclude that

$$\sigma \circ \rho = \tau$$

□

2.2 Ramification Theory

2.2.1 Complete discrete valued fields

Here we fix $(K, |\cdot|_K)$ a complete discretely valued field. We denote by v_K the associated valuation. Let L be a finite extension of K . We denote by $|\cdot| = |\cdot|_L$ the extension of $|\cdot|_K$ to L and $v = v_L$ the extension of v_K to L .

We assume L/K is *separable* (not always needed, but leads to easier proofs).

Lemma 48. *The valuation ring $\mathcal{O}_L$ of L is the integral closure of the valuation ring $\mathcal{O}_K$ of K in L .*

Dimostrazione. Since $\mathcal{O}_L$ is integrally closed, it contains the integral closure of $\mathcal{O}_K$ in L . Take $x \in \mathcal{O}_L$. Since $|\cdot|_L = (|\cdot|_K \circ N_{L/K})^{1/[L:K]}$. We have

$$1 \geq |N_{L/K}(x)|_K = \left| \prod_{\sigma \in \text{Hom}_K(L, K^{\text{sep}})} \sigma(x) \right|_K$$

Let Ω be an algebraic closure of K . Let $|\cdot|_\Omega$ be the extension of $|\cdot|_K$ to Ω . By uniqueness, for $\sigma \in \text{Hom}_K(L, \Omega)$, we have that $|\sigma(x)|_\Omega = |x|_\Omega$. So $N_{L/K}(x)|_K = |x|_L^{[L:K]}$, and so $|\sigma(x)|_L \leq 1$ for all $\sigma \in \text{Hom}_K(L, \Omega)$.

Using the relations of roots and coefficients, we deduce that the minimal polynomial μ_x of x is in $\mathcal{O}_K[X]$, so x is integral over $\mathcal{O}_K$. $\square$

In the following, we will use the following notations:

- $\mathcal{O}_K, \mathcal{O}_L$ valuation rings
- $\mathfrak{p}_K, \mathfrak{p}_L$ maximal ideals
- κ_K, κ_L the residue fields
- $e = e(L/K) = e(\mathfrak{p}_L|\mathfrak{p}_K)$ the ramification index: $\mathfrak{p}_K \mathcal{O}_L = \mathfrak{p}_L^e$
- $f = f(L/K) = f(\mathfrak{p}_L|\mathfrak{p}_K)$ the residual degree $f = [\kappa_L : \kappa_K]$

Osservazione. $e = [v_L(L^\times) : v_K(\kappa^\times)]$

And recall that $ef = [L : K]$, that L/K unramified means that κ_L/κ_K is separable and $e = 1$, and that L/K is totally ramified if $f = 1$

Lemma 49. *Take $a \in \mathcal{O}_L$. Assume that $f(a) = 0$ for some monic $f \in \mathcal{O}_K[X]$ and that $\bar{a} \in \kappa_L$ is a simple root of $\bar{f} \in \kappa_K[X]$. Then $\kappa(a)/\kappa$ is unramified, i.e.*

$$\kappa_{K(a)} = \kappa_K(\bar{a}) \quad \text{and} \quad [\kappa_{K(a)} : \kappa_K] = [K(a) : K]$$

Dimostrazione. Let $\bar{g} \in \kappa_K[X]$ be the minimal polynomial of $\bar{a}$. Let $g \in \mathcal{O}_K[X]$ be a monic lifting. Since $\bar{a}$ is a simple root of $\bar{f}$, it is also a simple root of $\bar{g}$. It is separable over κ_K , so $\kappa_K(\bar{a})/\kappa_K$ is separable, and by Hensel's lemma we can find $b \in \mathcal{O}_L$ such that $g(b) = 0$ and $\bar{b} = \bar{a}$.

$$\deg g = \deg \bar{g} \leq [\kappa_{K(b)} : \kappa] \leq [K(b) : K] \leq \deg g$$

So $[K(b) : K] = [\kappa_{K(b)} : \kappa_K]$. So $K(b)/K$ is unramified and $\kappa_{K(b)} = \kappa_K(\bar{a})$. Applying Hensel's lemma to f over $K(b)$, we can find a root $c \in \mathcal{O}_{K(b)}$ of f lifting $\bar{a}$. Such a root is necessarily unique. So we get that $c = a$, hence $a \in K(b)$.

Finally

$$\deg \bar{g} \stackrel{\bar{a} \in \kappa_{K(a)}}{\leq} [\kappa_{K(a)} : \kappa_K] \leq [K(a) : K] \leq [K(b) : K] = \deg g$$

Hence $K(a) = K(b)$ $\square$